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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2024

ME1007-1 – MATERIAL SCIENCE AND ENGINEERING
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which type of defect are point defects?
A) One dimensional defect
B) Zero dimensional defect
C) Two dimensional defect
D) Three dimensional defect
2. Burger vectors are relevant to which of the following crystalline defects?
A) Point defects
B) Line defects
C) Interfacial defects
D) Bulk defects
3. X-rays have larger wavelengths than which of the following?
A) Gamma rays
B) Beta rays
C) Microwave
D) Visible light
4. Which type of deformation occurs due to dislocation?
A) Slip plastic deformation
B) Elastic deformation
C) It does not show deformation
D) Slip elastic deformation
5. According to Hume Rothery's rules, size of atoms must not differ by more than _____.
A) 5%
B) 15%
C) 35%
D) 55%
6. A solution of exchange of impurities for solvent atoms is called a _____.
A) Interstitial solid solution
B) Substitutional solid solution
C) Saturated solution
D) Unsaturated solution
7. The maximum concentration of solute that can be added is defined as _____.
A) Solution limit
B) Solubility limit
C) Concentration
D) Degrees of freedom
8. The point at which two liquidus lines meet is known as _____.
A) Eutectic point
B) Isothermal point
C) Solvus point
D) Peritectic point
9. Steels containing 0.8% C are called _____.
A) Hypoeutectoid
B) Eutectoid
C) Hypereutectoid
D) Mild eutectoid
10. Hardenability of a material can be measured using _____ test.
A) Jominy end-quench
B) Charpy
C) Rockwell
D) Izod

11. Which of the following is applicable to Alpha iron?
 A) Soft B) Ductile
 C) Magnetic D) All of the mentioned
12. What does CCT diagram stand for?
 A) Constant-Critical-Temperature B) Constant-Cooling-Temperature
 C) Continuous-Cooling-Transformation D) Continuous-Creep-Transformation
13. How is cooling of the material done is Normalizing process?
 A) Furnace B) Cooling tower
 C) Still Air D) Liquid chamber
14. Process annealing is applied to which kind of materials?
 A) Steel castings B) Low carbon steels
 C) High carbon steels D) Medium carbon steels
15. Annealing a metal _____ its internal strains.
 A) doesn't affect B) removes
 C) increases D) doubles
16. In the process of hardening of steel, the metal has to be heated to a temperature _____ the critical range.
 A) below B) half
 C) above D) around
17. Which of the following is associated with minimum plastic deformation?
 A) Ductile fracture B) Brittle fracture
 C) Fatigue D) It doesn't occur during fracture
18. Below which point does fatigue occur?
 A) Ultimate strength B) Fracture point
 C) Elastic limit D) Yield point
19. Which of the following is a slow rise of plastic deformation under the action of stresses?
 A) Brittle fracture B) Ductile fracture
 C) Creep D) Fatigue
20. Which of the following may alter the mechanical properties of reinforced composites?
 A) Constituent properties B) Fiber length
 C) Fiber orientation D) All of the mentioned

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain types of Point Imperfections.	04	L*1	01	01
	b) Draw a neat sketch of edge and screw dislocations.	04	L1	01	01
	c) What is solidification? Explain solidification in metals.	04	L2	02	01
	d) Explain cast metal structure with neat sketch.	04	L1	02	01
2.	a) Define and give example.				
	i) Unit Cell	04	L1	01	01
	ii) Atomic Packing Factor	04	L1	01	01
	b) Give the detailed classification of engineering materials.				
	c) What is solid solution? Mention the types of solid solution. Explain the factors given by Hume Rothery that govern the formation of solid solution.	08	L2	02	01
3.	a) What you mean by surface defects? Explain different types of surface defects.	08	L2	01	01

- b) Two metals A and B have their melting points at 900°C and 800°C respectively. The alloy pair forms a eutectic at 600°C of composition 60% B and 40% A. A and B have unlimited mutual liquid solubilities. Their solid solubilities are as follows.
 10% B in A at 600°C and 5% B in A at 0°C
 8% A in B at 600°C and 4% A in B at 0°C
 Assume the liquidus, solidus and solvus lines to be straight. No solid state reactions or any intermediate phase changes occur in the series,

- i) Draw the phase diagram for the series and label all salient temperatures, compositions and regions.
- ii) Find the room temperature (25°C) structure of any alloy of composition 60% A and 40% B, with respect to the number, type, extent and composition of the phases.

08 L3 02 01

Unit – II

4. a) Draw a neat sketch of Fe–Fe₃C phase diagram and briefly explain. 08 L3 03 01
- b) Write a note on Surface treatments. Explain the Induction hardening process with neat sketch. 08 L2 04 01
5. a) What is the indication of T-T-T diagram? Explain with a neat diagram the Martensitic transformation of Austenite. 08 L2 03 01
- b) Explain Austempering and Annealing heat treatment processes with sketches. 08 L2 04 01
6. a) With neat sketch explain the process, how to find hardenability of a material using Jominy End-Quench Test? 08 L2 03 01
- b) Explain Martempering and Hardening heat treatment processes with sketches. 08 L2 04 01

Unit – III

7. a) Explain types of fracture with suitable example. 04 L1 05 01
- b) Explain with the sketch cup and cone fracture. 04 L1 05 01
- c) Briefly explain reinforcement materials used in composites. 04 L1 05 01
- d) Differentiate between Thermo setting and Thermoplastic polymers. 04 L1 05 01
8. a) Draw S-N curve, briefly explain. 04 L1 05 01
- b) Define Creep. With the help of sketch explain stages of Creep. 04 L1 05 01
- c) Write a note on Aluminum alloys and Copper Alloys properties, composition and list the applications. 08 L1 05 01

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
